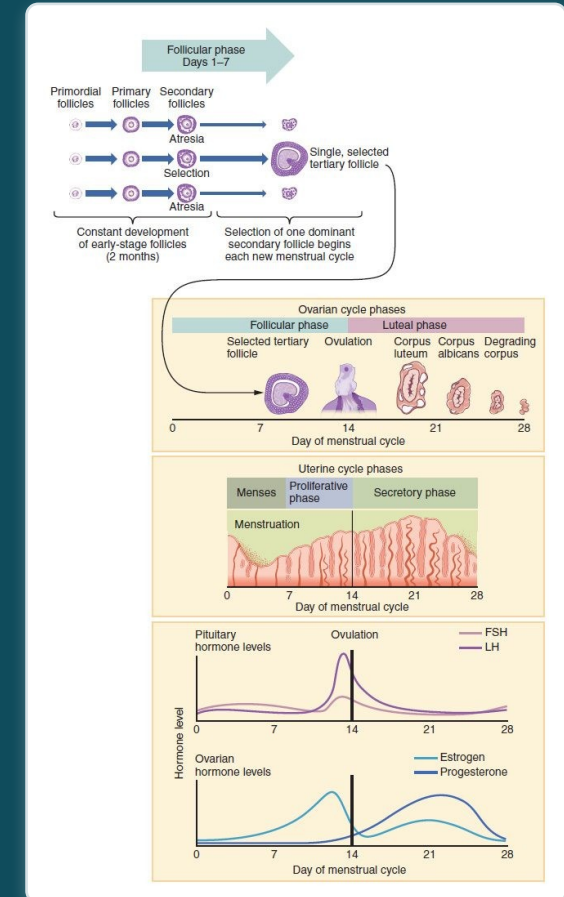


CHAPTER 27

The Reproductive System

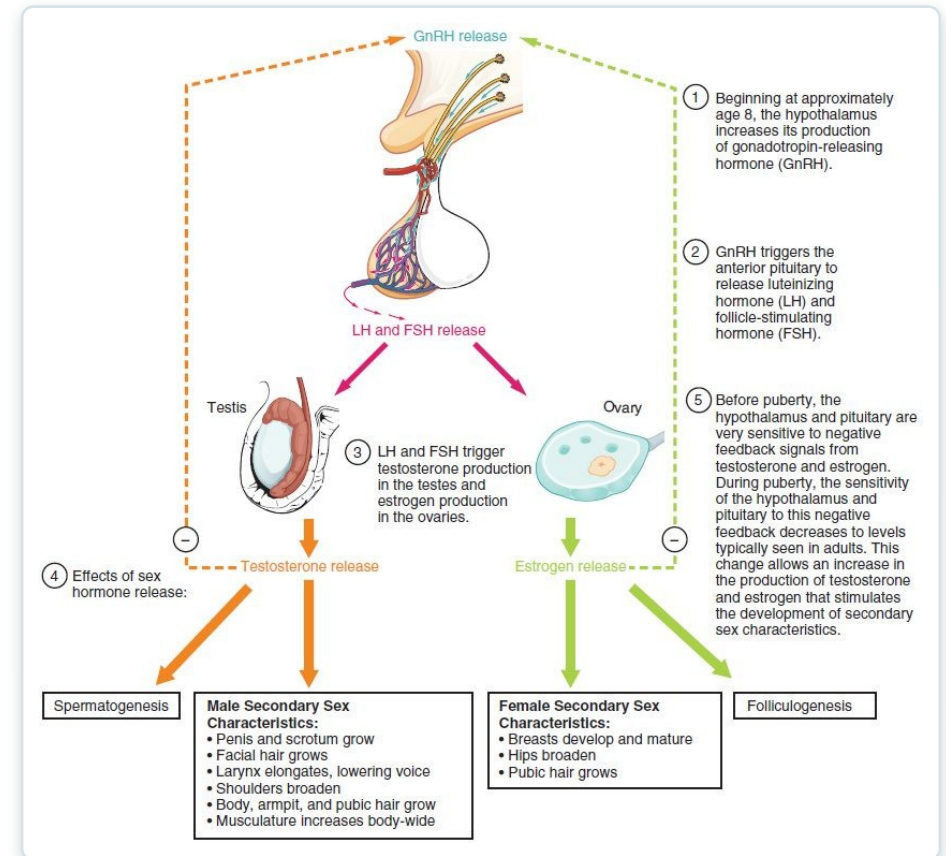
The organs of reproduction and how new life begins.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Describe the anatomy of the reproductive systems, including their accessory structures
- 2 Explain the role of hypothalamic and pituitary hormones in reproductive function
- 3 Trace the path of a sperm cell from its initial production through fertilization of an oocyte
- 4 Explain the events in the ovary prior to ovulation



Reproductive development and its hormones.



27.1

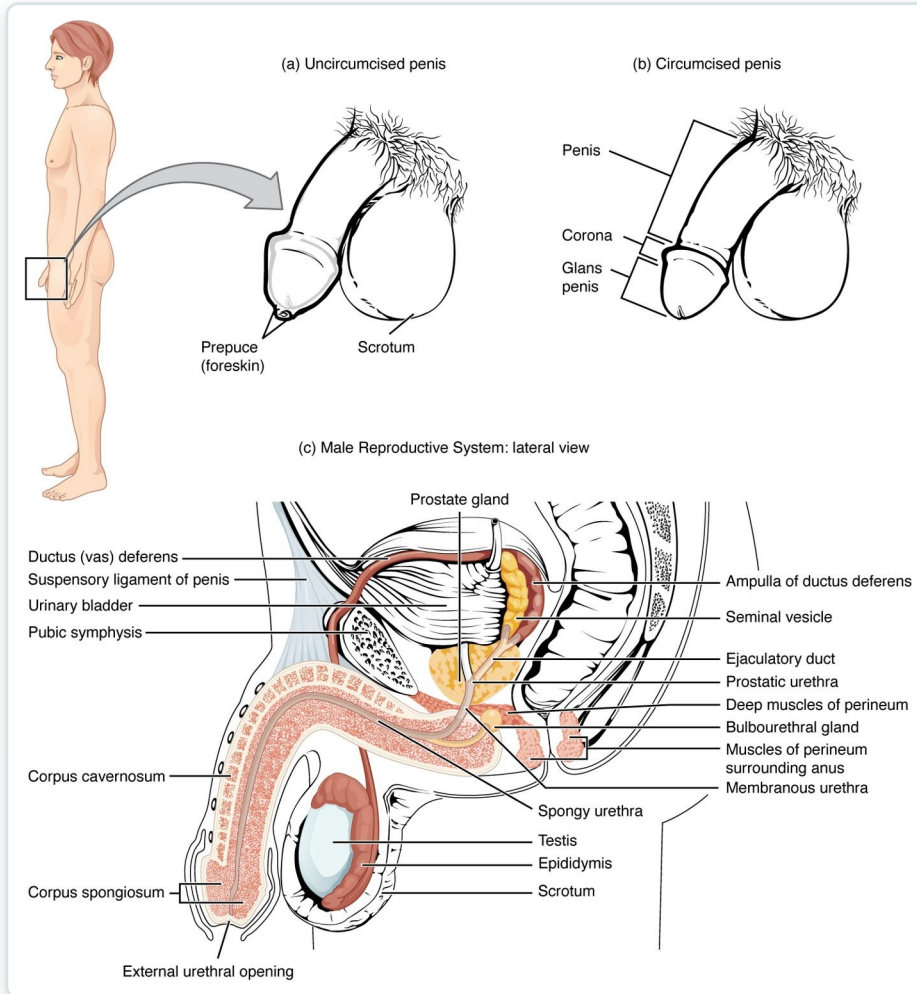
SECTION

Anatomy and Physiology of the Testicular Reproductive System

The male reproductive organs and sperm production.

SECTION 27.1

The Male Reproductive System



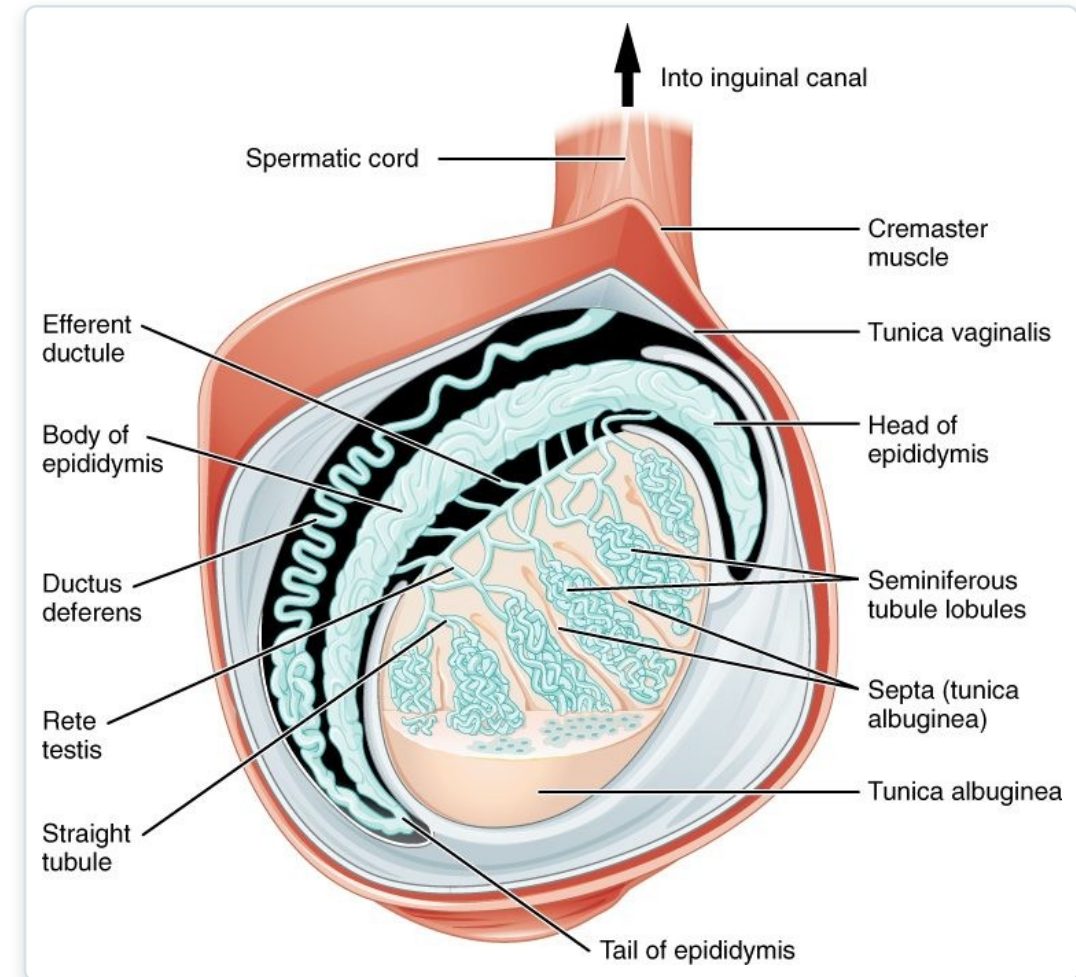
The male reproductive system

- Produces and delivers sperm
- Testes make sperm and testosterone
- Ducts carry sperm out
- Glands add seminal fluid

SECTION 27.1

The Testis & Epididymis

- The testes sit in the scrotum
- Sperm form in seminiferous tubules
- They mature in the epididymis
- The scrotum keeps them cool

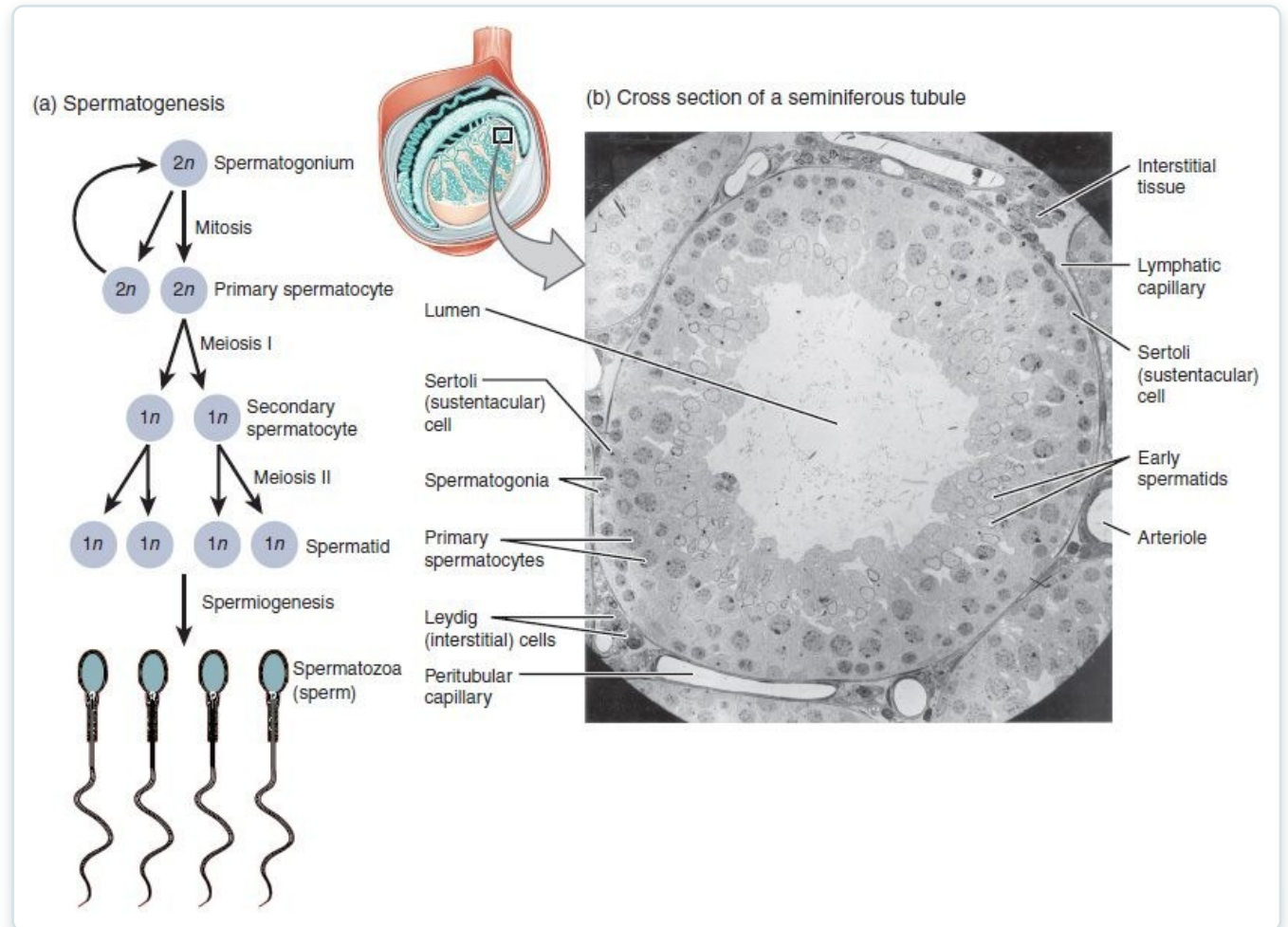


The testis and epididymis

SECTION 27.1

Spermatogenesis

- The making of sperm
- Begins at puberty
- Continues throughout life
- Each sperm carries half the chromosomes



Spermatogenesis



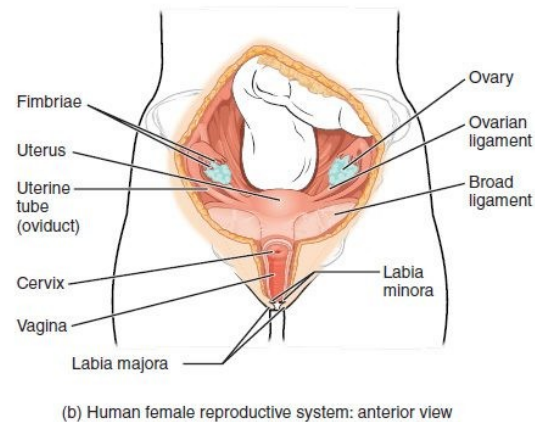
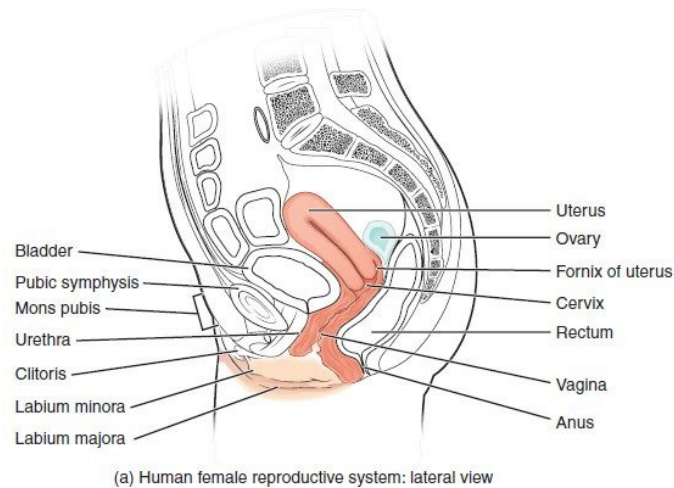
27.2

SECTION

Anatomy and Physiology of the Ovarian Reproductive System

The female reproductive organs, the egg, and the menstrual cycle.

The Female Reproductive System



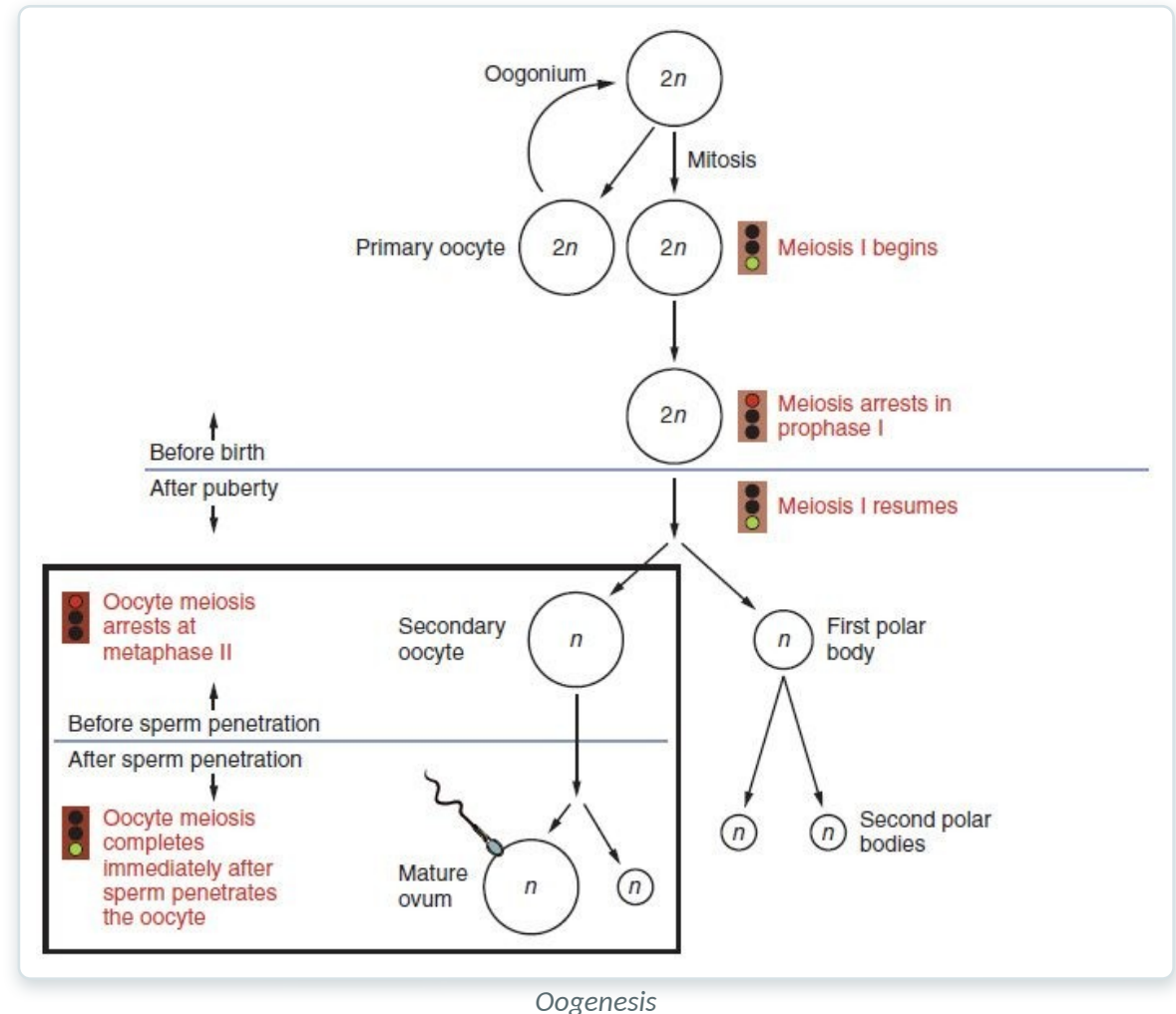
The female reproductive system

- Produces eggs and supports pregnancy
- Ovaries make eggs and hormones
- Uterine tubes carry the egg
- The uterus houses a pregnancy

SECTION 27.2

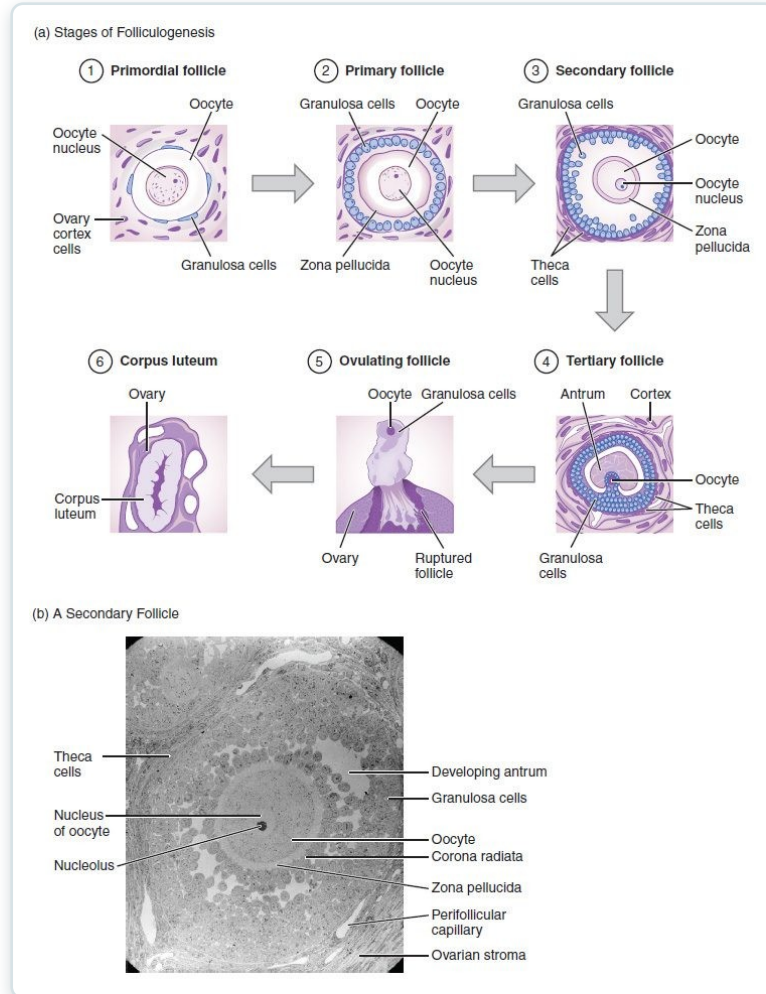
Oogenesis

- The making of eggs
- Begins before birth
- One egg matures each cycle
- Each egg carries half the chromosomes



SECTION 27.2

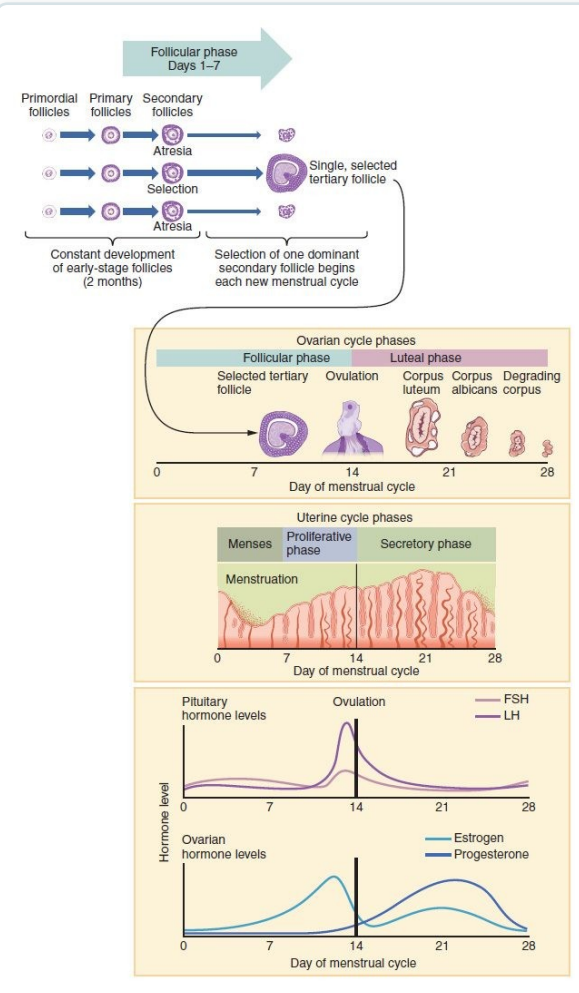
The Ovarian Cycle



Follicle development in the ovary

- Follicles develop in the ovary
- One matures and ovulates
- The empty follicle becomes the corpus luteum
- It secretes hormones

The Menstrual Cycle



The ovarian and menstrual cycle

- A roughly monthly cycle
- Hormones drive ovarian and uterine changes
- The uterine lining builds up
- It sheds as menstruation if no pregnancy



27.3

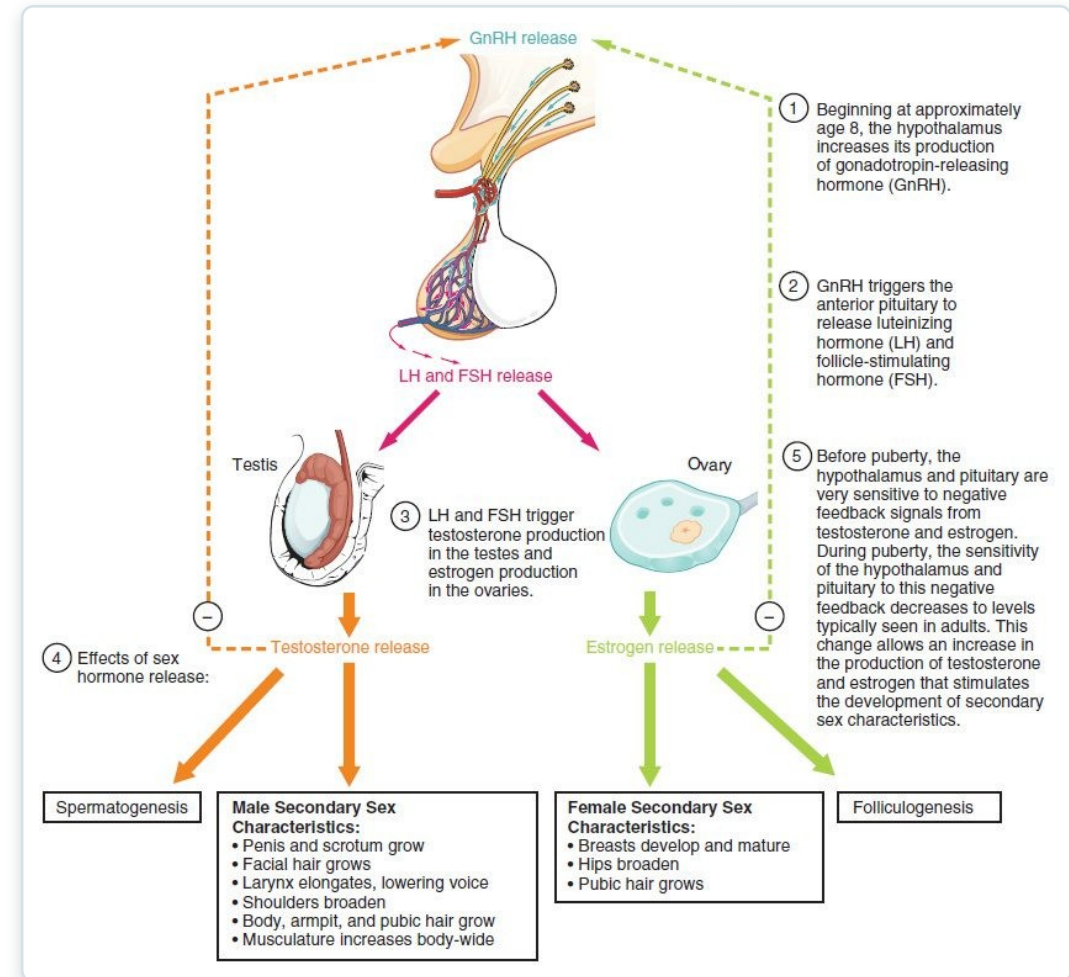
SECTION

Development of the Male and Female Reproductive Systems

How the reproductive systems develop and mature at puberty.

Reproductive Development & Puberty

- Sex is determined at fertilization
- Systems develop before birth
- They mature at puberty
- Driven by reproductive hormones



Reproductive development at puberty

Key Takeaways

- The male system produces and delivers sperm; the female system produces eggs and supports pregnancy.
- Sperm and eggs each carry half the chromosomes, restored to a full set at fertilization.
- The testes make sperm and testosterone; the ovaries make eggs, estrogen, and progesterone.
- The menstrual cycle coordinates ovulation with the uterine lining each month.
- Reproductive organs mature at puberty under control of the pituitary hormones.

CLINICAL CONNECTIONS

Why the Reproductive System Matters in Practical Nursing

Contraception

Methods work at different points in reproduction.

STIs

Sexually transmitted infections require prevention and treatment.

Menstrual Disorders

Irregularities can signal hormonal or other issues.

Menopause

The natural end of the reproductive years.

Prostate Health

The prostate commonly enlarges with age in men.

Fertility

Many factors affect the ability to conceive.